

ADMINISTRATION MANUAL

RazakEvent KTR Admin Guide

Comprehensive guide for KTR administrators and event organizers covering event creation, QR code generation, participant management, report uploads, and system administration.

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Table of Contents

1. Introduction	2
2. Admin Access and Login	2
3. Event Management	3
3.1 Creating a New Event	3
3.2 Editing and Managing Events	4
3.3 Event Approval System	5
4. QR Code Generation	5
4.1 Generating a QR Code for Your Event	5
4.2 Best Practices for QR Code Display	5
5. Participant Management	6
5.1 Viewing the Participant List	6
6. Post-Event Report Upload	6
6.1 Uploading a Report	7
6.2 Report Content Guidelines	7
7. Event Data and Analytics Overview	7
8. Troubleshooting for Admins	8
9. Technical Architecture Reference	9

1. Introduction

This manual is intended for Kolej Teknologi Razak (KTR) administrators and event organizers who are responsible for creating, managing, and overseeing events within the RazakEvent platform. As an admin user, you have elevated privileges that allow you to create new events, generate QR codes for attendance tracking, view participant lists, manage event approvals, and upload post-event reports. These tools are designed to streamline the entire event lifecycle from creation to completion reporting.

RazakEvent is built on a modern technology stack using Flutter for the mobile frontend and Firebase for the backend services including authentication, database (Firestore), and file storage. The application follows the MVVM (Model-View-ViewModel) architecture pattern which ensures clean separation of concerns and maintainability. Admin features are accessible through a dedicated Organizer tab within the app, which appears only for users with admin or organizer role assignments.

This guide covers each admin feature in detail with step-by-step instructions. It is organized by functional area so you can quickly find the information you need. Whether you are setting up a new event, checking attendance records, or generating compliance reports, this manual will walk you through the process clearly and comprehensively.

2. Admin Access and Login

Admin and organizer accounts are created by the system administrator and assigned specific roles within the Firebase authentication system. Unlike regular student accounts, admin accounts have additional permissions that unlock the Organizer section of the app. If you believe you should have admin access but do not see the Organizer tab, please contact the system administrator to verify your role assignment.

The login process for admin users is identical to the student login process. You will use your assigned UTM email address and password to sign in through the same login screen. Once authenticated, the system checks your user role and dynamically displays the Organizer tab in the bottom navigation bar if you have been granted admin or organizer privileges. This role-based access control ensures that only authorized personnel can create and manage events within the system.

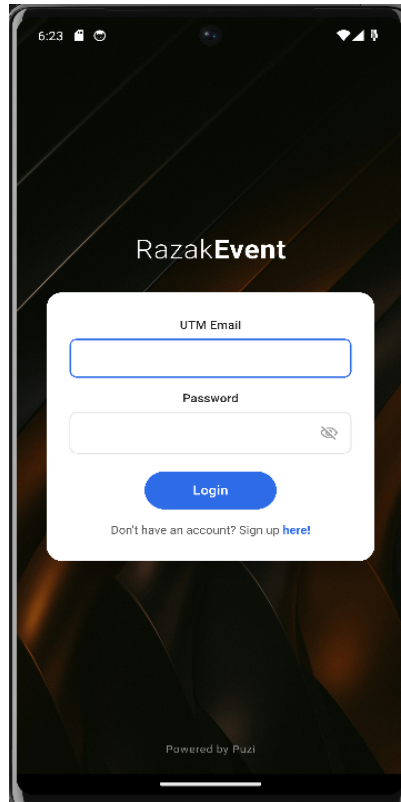


Figure 1: Admin Login Screen (same interface for all users)

Role	Permissions
Admin	Full access: create events, approve events, manage users, generate QR codes, view all reports, upload documents, and access system settings.
Organizer	Limited access: create and manage own events, generate QR codes for own events, view participant lists, and upload event reports.
Student	Basic access: browse events, scan QR codes for attendance, and view own profile and leaderboard standings.

Table 1: User Roles and Permissions

3. Event Management

Event management is the primary responsibility of KTR administrators and organizers. The RazakEvent platform provides a comprehensive set of tools for creating, editing, and managing events throughout their lifecycle. This section covers the complete workflow from creating a new event to managing its status and visibility within the app.

3.1 Creating a New Event

To create a new event, navigate to the Organizer tab in the bottom navigation bar. This tab is only visible to users with admin or organizer role assignments. Once in the Organizer section, you will find the event creation form which allows you to define all the details of your event. Follow these steps to create a new event:

1. Tap the "Organizer" tab in the bottom navigation bar.
2. Tap the "Create Event" button to open the event creation form.
3. Fill in the event details including:
 - Event Title: A clear, descriptive name for the event (e.g., "Python Workshop: Introduction to Data Analysis")
 - Description: A detailed description of what the event is about, its objectives, and what participants can expect to learn or experience.
 - Date and Time: Select the start date, start time, end date, and end time for the event using the date and time pickers.
 - Venue/Location: Specify where the event will take place (e.g., "Lab DK 5, Kolej Teknologi Razak").
 - Category: Select the most appropriate category from the available options such as Academic, Sports, Cultural, Technology, or Workshop.
 - Maximum Participants: Set the maximum number of participants allowed (optional).
4. Review all the information for accuracy.
5. Tap "Create" or "Submit" to publish the event.

After submission, the event will appear in the system and will be visible to students on the Home screen. Depending on the approval workflow configured for your organization, the event may need to be approved by a higher-level admin before it becomes visible to all users.

3.2 Editing and Managing Events

As an organizer or admin, you can edit events that you have created (or all events if you have admin privileges). To manage an existing event, navigate to the Organizer tab and find the event in your event list. You can update any of the event details including the title, description, date, time, venue, and category. Changes are saved immediately to the Firebase database and will be reflected on the student-facing Home screen in real time.

You can also change the status of an event. Common status transitions include marking an event as "Cancelled" if it needs to be called off, or "Completed" after the event has concluded. These status changes help keep the event list accurate and inform students about the current state of each event. It is good practice to update event statuses promptly so that students always see up-to-date information.

3.3 Event Approval System

The event approval system is a quality control mechanism that ensures only appropriate and well-organized events are published to the student-facing app. When an organizer creates a new event, it may be placed in a "Pending Approval" state until an admin reviews and approves it. Admins can access the approval queue from the Organizer section, where they can review event details and either approve or reject each pending event.

Approved events immediately become visible to all students on the Home screen. Rejected events are sent back to the organizer with feedback explaining the reason for rejection. This two-tier system helps maintain the quality and relevance of events published on the platform, preventing incomplete or inappropriate event listings from reaching students.

4. QR Code Generation

QR codes are the primary mechanism for tracking student attendance at events. Each event has a unique QR code that students scan using the Scan feature in their app. As an organizer, you are responsible for generating the QR code for your event and displaying it at the event venue so that students can scan it to record their attendance.

4.1 Generating a QR Code for Your Event

To generate a QR code for a specific event, follow these steps:

1. Navigate to the Organizer tab in the bottom navigation bar.
2. Find the event for which you want to generate a QR code in your event list.
3. Tap on the event to open its details page.
4. Look for the "Generate QR Code" or "QR Code" button.
5. Tap the button to generate the unique QR code for the event.
6. The QR code will be displayed on screen. You can save it as an image or display it directly from your device at the event venue.

Each QR code is cryptographically unique and tied to a specific event in the Firebase database. When a student scans the QR code, the system validates the code against the database and records the attendance entry with a timestamp. This ensures that attendance records are accurate and tamper-proof.

4.2 Best Practices for QR Code Display

To ensure smooth attendance scanning at your event, follow these best practices when displaying the QR code at the venue:

- Print the QR code in a large size (at least 15 cm x 15 cm) on paper or foam board.
- Place the QR code at eye level at the entrance or registration desk.
- Ensure the venue has adequate lighting near the QR code display area.
- Have a backup printed copy in case the primary one gets damaged.
- If displaying on a screen, ensure the screen brightness is set to maximum.
- Station a volunteer near the QR code to assist students who have difficulty scanning.

5. Participant Management

As an event organizer or admin, you can view the list of participants who have scanned the QR code and recorded their attendance at your event. The participant management feature provides real-time visibility into attendance numbers and participant details, which is essential for event reporting and compliance purposes.

5.1 Viewing the Participant List

To view the participant list for an event, navigate to the Organizer tab, select the event, and look for the "Participants" or "Attendance" section. The participant list shows each student who has scanned the event QR code, along with their name, faculty, and the exact timestamp of their scan. This information is pulled directly from the Firebase database in real time, so the list updates automatically as more students scan the QR code during the event.

You can use the participant list to verify attendance numbers, identify students who have not checked in, and generate attendance reports for your records. The list can be reviewed during the event to monitor check-in progress or after the event for final attendance records.

6. Post-Event Report Upload

After an event has concluded, organizers are typically required to submit a post-event report documenting the outcomes, attendance figures, feedback, and any issues encountered during the event. The RazakEvent platform includes a report upload feature that allows organizers to attach documents directly to the event record in the system, making it easy to maintain a complete archive of all event documentation.

6.1 Uploading a Report

To upload a post-event report, follow these steps:

1. Navigate to the Organizer tab and select the completed event.
2. Look for the "Upload Report" or "Attachments" section in the event details.
3. Tap the upload button to open the file picker.
4. Select the report file from your device (supported formats include PDF, DOCX, and common image formats).
5. Wait for the upload to complete. The file will be stored in Firebase Storage.
6. Verify that the uploaded file appears in the event attachments list.

Note: Uploaded reports are stored in Firebase Storage and are accessible to admins and the event organizer. Ensure that all reports are uploaded in a timely manner after the event conclusion to maintain an up-to-date event archive.

6.2 Report Content Guidelines

A comprehensive post-event report should include the following information to ensure complete documentation of the event for future reference and organizational reporting:

Section	Description
Event Summary	Brief overview of the event including title, date, time, venue, and organizer name.
Objectives	The goals and learning outcomes that the event aimed to achieve.
Attendance Data	Total number of participants, expected vs. actual attendance, and attendance percentage.
Activity Details	Description of activities, sessions, or workshops conducted during the event.
Feedback Summary	Key feedback received from participants, satisfaction ratings, and notable comments.
Issues Encountered	Any problems, delays, or challenges faced during the event and how they were resolved.
Recommendations	Suggestions for improving similar events in the future based on lessons learned.

Table 2: Post-Event Report Sections

7. Event Data and Analytics Overview

The RazakEvent platform stores all event and attendance data in Firebase Firestore, which provides real-time data synchronization and querying capabilities. As an admin, understanding the data model helps you manage events more effectively and generate accurate reports. The following table describes the key data models used in the system and how they relate to each other in the MVVM architecture.

Model	Purpose	Key Fields
Event	Stores event details and metadata	title, description, date, time, venue, category, organizerId, status
User	Stores user profile and role information	name, email, faculty, role (student/organizer/admin), meritPoints
Attendance	Records QR scan attendance entries	eventId, userId, scannedAt (timestamp), status
Report	Links uploaded documents to events	eventId, fileUrl, uploadedAt, uploadedBy

Table 3: Core Data Models

8. Troubleshooting for Admins

This section covers common issues that administrators and organizers may encounter while using the RazakEvent platform. For issues related to the student-facing features, refer to the Student User Manual. If you encounter a problem not listed here, please contact the technical support team or the lead developer through the project repository.

Issue	Solution
Organizer tab not visible	Verify your user role in Firebase Auth. Contact the system administrator to ensure your account has been assigned the organizer or admin role. Try logging out and back in.
Cannot create events	Check your internet connection. Ensure you have organizer or admin privileges. Verify that all required fields in the event form are filled correctly.
QR code not generating	Ensure the event has been saved successfully before generating the QR code. Check your connection to Firebase. Try refreshing the event details page.
Participant list not updating	Pull down to refresh the list. Check your internet connection. Note that there may be a slight delay (1-2 seconds) between a student scanning and the list updating.
Report upload failing	Check the file size (large files may timeout on slow connections). Ensure the file format is supported (PDF, DOCX, JPG, PNG). Verify Firebase Storage permissions.
Event not visible to students	The event may be in "Pending Approval" status. An admin needs to approve the event. Also verify the event date is current or in the future.

Table 4: Admin Troubleshooting Guide

9. Technical Architecture Reference

For administrators who need to understand the technical underpinnings of RazakEvent, this section provides a brief overview of the application architecture. RazakEvent follows the MVVM (Model-View-ViewModel) pattern with Provider for state management. The data flow proceeds from Views through ViewModels to Repositories and finally to Firebase services. This architecture ensures that the UI remains responsive while data operations happen asynchronously in the background.

The project currently consists of 20 views, 6 models, 5 repositories, and 13 ViewModels. Firebase provides three core services: Firebase Authentication for user login and role management, Cloud Firestore for real-time database operations including event data and attendance records, and Firebase Storage for file uploads such as post-event reports and event images. All network communication is handled asynchronously using Dart futures and streams, ensuring a smooth user experience even on slower network connections.

Component	Technology
Framework	Flutter (Dart)
Architecture	MVVM (Model-View-ViewModel)
State Management	Provider
Authentication	Firebase Authentication
Database	Cloud Firestore
File Storage	Firebase Storage
QR Scanning	mobile_scanner package

Table 5: Technology Stack